VAPI Gamer Foundation Guide

What VAPI Actually Does for You — The Honest, Complete Guide

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Classification: Consumer-Facing Technical Guide

Audience: Competitive gamers, esports participants, casual players considering VAPI enrollment, gaming community leaders

“Written for gamers, by the protocol team. No marketing spin.”

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1. What This Guide Is (And What It Isn’t)

This document is a transparent, technically grounded explanation of VAPI — Verified Autonomous Physical Intelligence — written entirely from the competitive player’s perspective. It is here to answer one question honestly: **what does VAPI actually do for you?**

This guide covers:

- What VAPI protects you from and how it works at a technical level
- What VAPI **cannot** do yet — stated plainly, without spin
- How your biometric and controller data is handled, stored, and destroyed
- What enrollment means, what it costs, and what you get from it
- What your rights are over your own data

ⓘ **HONEST NOTE**

This is **not** a sales pitch. VAPI has real limitations right now, and every one of them is disclosed in Section 7. Where the technology is still in development, we say so. Where calibration data is insufficient, we give you the actual numbers. If you're evaluating VAPI, you deserve the full picture — not a highlight reel.

If you're a competitive player who is tired of losing to cheaters that current anti-cheat systems physically cannot detect, or if you care about being able to *prove* that your skill is real, this guide tells you everything you need to know to make an informed decision.

2. The Problem VAPI Solves

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Why Existing Anti-Cheat Fails You

Every major anti-cheat system in competitive gaming today — Valve Anti-Cheat (VAC), Easy Anti-Cheat (EAC), BattlEye, Riot's Vanguard, FACEIT Anti-Cheat — works on the same fundamental principle: they scan your PC's memory, running processes, and loaded drivers, looking for known cheat software signatures or suspicious memory patterns.

This approach has served the industry for years. But it has **fundamental blind spots** that no amount of software scanning can fix — and those blind spots directly hurt legitimate players.

2.1 Hardware Injection Devices

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The Invisible Cheat

Devices like the **Cronus Zen** and **Titan One** represent the single biggest unsolved problem in competitive gaming anti-cheat. Here is how they work:

- The device sits physically between a legitimate controller and the PC, connected via USB relay
- It intercepts controller signals and injects scripted inputs — recoil compensation, rapid fire sequences, aim assist scripts — through the clean controller
- The PC sees a legitimate controller sending legitimate-looking HID input reports — because as far as the USB protocol is concerned, that is exactly what is happening
- There is **no software cheat to detect**. No rogue process. No injected DLL. No modified memory. The cheat operates entirely below the software layer.

The result: **a player using a $40 hardware device can defeat million-dollar anti-cheat systems**. Every existing anti-cheat — VAC, EAC, BattlEye, Vanguard, FACEIT — is completely blind to these devices because they only scan software. This is not a criticism of those systems; it is a structural limitation of software-only detection.

2.2 Identity Fraud

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Boosting, Smurfing, and Carry Services

Current anti-cheat can tell you what software is running on a machine. It **cannot tell you who is holding the controller.**

- **Account boosting:** A highly skilled player logs into your account and plays on your behalf, inflating your rank far beyond your actual skill level
- **Smurfing:** A high-ranked player creates a new account with a deliberately low rank, then dominates lobbies full of significantly weaker players
- **Carry-for-hire:** Paid services where someone else plays ranked matches on your behalf for money

These practices corrupt rankings, destroy matchmaking quality, and undermine competitive integrity. When a Diamond-level player is boosting a Bronze account through Platinum lobbies, every legitimate Platinum player in those matches is affected. Current systems have no solution because they have no concept of player identity at the physical level.

2.3 Macro and Script Automation

Keyboard and controller macros replay scripted input sequences with inhuman precision. A human being cannot press a series of buttons with microsecond-level timing consistency — but a macro can, and does, every single time.

Software-level macro detection catches some of these by analyzing input timing patterns at the application layer. But when macros are replayed through a **hardware injection device** (see Section 2.1), the inputs arrive via the USB HID protocol from what appears to be a clean controller. There is no software hook to detect. The macro is invisible to every existing anti-cheat system.

2.4 The Kernel-Level Privacy Problem

To fight increasingly sophisticated cheats, some anti-cheat systems have escalated to **kernel-level ring-0 drivers**. Riot's Vanguard and certain configurations of EAC install drivers that run at the deepest level of your operating system — the kernel. These drivers have complete, unrestricted access to everything on your system, and they run continuously, not just when you're playing.

This creates legitimate security and privacy concerns:

- A vulnerability in a ring-0 driver is a vulnerability in your *entire system* — it runs with the same privileges as your operating system kernel

- The driver can see all running processes, all memory, all file operations — always
- Players are forced into an impossible choice: accept deep system access to play competitively, or refuse and be locked out of ranked modes

ⓘ **KEY DIFFERENCE**

VAPI does NOT require kernel-level access. The VAPI bridge runs entirely in user space on your PC. No ring-0 drivers. No system-wide hooks. No always-on background processes scanning your entire system. It reads controller data via the standard USB HID interface — the same data path your game already uses.

3. How VAPI Actually Works

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The Honest Technical Explanation

VAPI operates at the **physics layer**. Instead of scanning your PC's software looking for cheat signatures, it reads the physical signals your body produces when you hold and use a game controller. This section explains the technology plainly, without hiding complexity or oversimplifying the science.

3.1 What VAPI Reads From Your Controller

When you hold a game controller, your body generates a continuous stream of involuntary physical signals. You are not aware of most of them, but they are measurable, consistent, and unique to you. Here is what VAPI reads:

Physical Signal	What It Means
Neurological tremor (8–12 Hz)	Your hands produce tiny involuntary vibrations driven by your nervous system. The frequency and amplitude of this tremor are unique to you — like a fingerprint, but from your neuromuscular system.
Trigger onset velocity	When you press a trigger, your finger has a characteristic speed and smoothness profile. How quickly you initiate a press, how the pressure ramps up — this varies between individuals but is remarkably consistent for each person.
Grip asymmetry	Your left and right hands grip the controller differently. The pressure distribution, the micro-movements, the steadiness — VAPI measures the asymmetry between your two hands.
Stick autocorrelation patterns	When you move an analog stick, human movement follows characteristic smoothness curves. Your stick input at any given moment is statistically correlated with your input moments before. Bots and scripts do not produce this natural autocorrelation.
Pre-press IMU movement	The micro-accelerometer in your controller detects that your hand muscles physically shift the controller slightly *before* every button press. A human being cannot press a button without first moving the controller, even imperceptibly. A hardware injection device injects button presses **without** any preceding physical movement.

All of this data is read at **1,000 times per second** (1,000 Hz over USB) from a standard controller. The reference device is the Sony DualShock Edge (CFI-ZCP1).

3.2 The 228-Byte PoAC Record

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Your Proof of Play

Every approximately **1 millisecond**, VAPI assembles a compact **228-byte** data record called a **Proof of Autonomous Cognition (PoAC)**. This is the fundamental unit of evidence in the VAPI system. Each record contains:

Field	Description
Timestamp	Nanosecond-precision timestamp of when the record was generated
Inference code	What VAPI detected at this moment: normal play, an anomaly, or a specific cheat type
Confidence score	How confident the system is in its inference
Session & sequence numbers	Identifies your session and the exact position of this record in the chain
Biometric anomaly score	How far your current input deviates from your established biometric profile
Continuity score	Whether the play pattern is continuous and consistent, or shows signs of interruption/substitution
12 biometric features	Statistical features extracted from controller physics (tremor, grip, timing — see Section 4.1)
Device unique ID	Cryptographic identifier of your specific controller
ECDSA-P256 signature	A cryptographic signature proving this record came from your specific controller and was not forged

The body of each record (the first **164 bytes**) is hashed with **SHA-256** and linked to the hash of the previous record — forming an unbreakable chain. If anyone tampers with any single record, the hash chain breaks and the tampering is immediately detectable. This is the same chaining principle used in blockchain technology, applied at the individual record level.

The cryptographic signature proves the record originated from your specific controller's secure element. It cannot be forged without physical possession of your device.

> ⓘ **WHAT THIS MEANS IN PRACTICE**
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> A PoAC chain is **cryptographic proof that YOU were physically playing**. Not just that your account was active. Not just that your PC was running clean software. But that a human being with *your specific biometric profile* was holding the controller, producing real physical signals, at every millisecond of your session.

3.3 The Nine Detection Layers (PITL Stack)

VAPI's detection system is organized into nine layers, each targeting a different type of cheating or verification. Together they form the Physical Intelligence Trust Layer (PITL) stack. Here is what each layer does, in plain language:

Layer	Name	What It Does
L0	Device Presence	Is a real controller connected? Is it polling at the correct rate (**800–1,100 Hz**)? This is the baseline check — if no legitimate device is present, nothing else matters.
L1	Chain Integrity	Is the PoAC chain unbroken? No gaps in the sequence, no replayed old records, no timestamp reversals. If anyone tries to splice in fake records or replay a previous session, L1

Layer	Name	What It Does
		catches it.
L2	Hardware Injection Detection	**This is the big one.** VAPI reads the controller's IMU (accelerometer/gyroscope) and compares it to the HID input report. A Cronus Zen injects button presses *without physically moving the controller*. VAPI detects this discrepancy — a button press with no corresponding IMU movement is physically impossible for a human. When L2 fires (inference code `0x28`), it means a hardware injection device has been detected. **This is something no other anti-cheat system can do.**
L3	Behavioral Cheat Detection	TinyML classifiers trained on behavioral signatures of aimbots and wallhacks. When someone's aiming pattern matches known bot behavior (e.g., inhuman snap-to-target trajectories, pre-aim through walls), L3 fires. Inference code `0x29` = wallhack detected; `0x2A` = aimbot detected.
L2B	IMU-Button Causal Latency	When a human presses a button, there is always a tiny preceding physical movement of the controller detected by the IMU — your muscles shift the device before the button actuates. Hardware injection produces button events with **no preceding movement**. L2B measures this causal

Layer	Name	What It Does
		latency gap.
L2C	Stick-IMU Cross-Correlation	When you move an analog stick, the controller physically tilts in your hand. VAPI correlates stick deflection with IMU tilt data. Bots move sticks without moving controllers — the cross-correlation drops to near zero.
L4	Biometric Fingerprint	Your personal 12-feature biometric profile. VAPI builds a statistical model of YOUR playing style — tremor frequency, trigger velocity, grip asymmetry. If a different person picks up your controller, L4 detects the biometric difference via Mahalanobis distance from your established centroid.
L5	Temporal Rhythm Oracle	Human button timing has natural **jitter** — slight, random variations in timing between presses. Macro scripts press buttons with zero jitter, producing impossibly consistent inter-press intervals. L5 measures timing entropy and flags sub-human consistency.
L6	Active Haptic Challenge	Sub-perceptual vibrations sent through the controller. Your body responds involuntarily within 80–280 ms via reflex arc. Bots do not respond. **Currently disabled** — needs N≥50 calibration sessions. See Section 7.2.

3.4 What Happens When VAPI Detects Something

Not all detections are equal. VAPI distinguishes between **hard cheat codes** (definitive physics violations) and **advisory codes** (statistical anomalies that contribute to an overall score).

Hard Cheat Codes — Unconditional Block:

Code	Detection	What It Means
`0x28`	Hardware injection (L2)	The IMU data physically contradicts the input data. A button was pressed but the controller did not move. This is not a probability judgment — it is a physics measurement.
`0x29`	Wallhack (L3)	Aim/movement patterns show consistent pre-aim toward targets through solid geometry with no line of sight.
`0x2A`	Aimbot (L3)	Aim trajectory shows inhuman snap-to-target kinematics inconsistent with human motor control.

These codes **block tournament eligibility unconditionally**. There is no appeal process for the detection itself because the underlying measurement is physical, not probabilistic.

Advisory Codes — Weighted Humanity Score:

Advisory flags (biometric anomaly, timing anomaly, IMU-button decoupling) contribute to a weighted **humanity_probability** score. A single advisory flag does **not** get you banned or blocked. The system uses this formula:

```
humanity_probability = 0.28

×

L4 + 0.27

×

L5 + 0.20

×

E4 + 0.15

×

L2B + 0.10

×

L2C
```

Multiple advisory signals are evaluated by **36 AI agents** running simultaneously, then verified through a decentralized consensus network called **ioSwarm**, where multiple independent nodes must agree before any action is taken.

The epistemic consensus threshold is **0.65** — and it is **deliberately designed so that no single agent or pair of agents can reach this threshold alone**. This prevents any one component from unfairly flagging a player. The architecture requires broad agreement across independent evaluators before any adverse action can be considered.

4. Your Biometric Data

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What VAPI Knows, What It Doesn

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t, And What It Can Never Do

This is the section most gamers care about — and the one where total transparency matters most. Your body produces the data that makes VAPI work. You deserve to know exactly what happens with it.

4.1 What VAPI Collects

VAPI reads **controller physics** — the raw sensor data from your controller's accelerometer, gyroscope, triggers, sticks, buttons, and touchpad. From these raw signals, it extracts **13 statistical features**:

#	Feature	What It Measures
1	Tremor variance	The intensity and consistency of your involuntary hand tremor
2	Trigger onset velocity (left)	How fast and smoothly you press the left trigger
3	Trigger onset velocity (right)	How fast and smoothly you press the right trigger
4	Grip asymmetry	The measurable difference between your left and right hand grip behavior
5	Stick autocorrelation (lag 1)	How your stick movement at one moment predicts your movement at the next
6	Stick autocorrelation (lag 2)	Same measurement over a slightly longer time window
7	Tremor peak frequency	The dominant frequency

#	Feature	What It Measures
		of your hand tremor (typically 8–12 Hz)
8	Tremor band power	How much energy is in the tremor frequency band
9	Spectral entropy	How “organized” or “random” your tremor spectrum is
10	Touchpad variance	The variability in your touchpad contact patterns
11	Timing jitter	The natural randomness in your button press timing
12	Touchpad spatial entropy	How your touchpad contact points distribute spatially
13	IMU movement signature	Your characteristic controller movement pattern during gameplay

These are **statistical derivations**. They describe mathematical patterns in your movement. They are not images of your body, not biological samples, and not personally identifying information in the traditional sense.

4.2 What VAPI Does NOT Collect

To be absolutely clear about what VAPI **never touches**:

- **No fingerprints.** No facial recognition. No retinal scans. No DNA. No biological samples of any kind.
- **No screen recording.** VAPI does not see your screen, capture screenshots, or record video.
- **No keystroke logging** beyond controller inputs. Your keyboard, your browser, your documents — untouched.

- **No file system access.** VAPI does not read, write, or scan any files on your PC.
- **No microphone access.** No camera access. No personal files.
- **No kernel-level access.** No ring-0 drivers. VAPI runs entirely in user space.

4.3 Seven Privacy Protections

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In Plain Language

VAPI implements seven specific biometric privacy primitives, designated BP-001 through BP-007. Here is what each one does, explained without jargon:

1. Your data expires automatically (BP-001 — Temporal Biometric Decay)

Your biometric profile has a **90-day half-life**. After 90 days, your old biometric data is mathematically weighted at 50% of its original value. After 180 days, it is effectively zero. VAPI does not hoard your data forever — it decays by design. Even if someone somehow obtained your old biometric data, it would be useless within months. Your profile is continuously rebuilt from recent sessions, not accumulated indefinitely.

2. Your consent is cryptographically proven (BP-002 — ZK-Attested Consent)

When you consent to biometric processing, it is recorded as a **zero-knowledge proof** — a mathematical proof that you consented without revealing your identity. Your consent is verifiable by anyone who needs to check it, but your identity stays hidden in the consent record itself. You can also **revoke consent** via ZK proof at any time, and the revocation is equally verifiable and privacy-preserving.

3. Mathematical noise protects your privacy (BP-003 — Differential Privacy)

Calibrated noise is mathematically added to your biometric thresholds. This means your individual data **cannot be reverse-engineered** from the processed results. You have a privacy budget of **ε ≤ 1.0 per year** — when this budget is exhausted, no more queries on your data are allowed until it resets. This is the same privacy standard used by the US Census Bureau for protecting individual respondent data.

4. You're hidden in a crowd (BP-004 — K-Anonymity)

Your biometric thresholds are only activated when your cohort — players with the same controller type, transport type, and similar play patterns — has **at least 5 members**. You are statistically indistinguishable from others in your cohort. Your individual biometric profile cannot be singled out from the group. If your cohort is too small, the thresholds are not applied, protecting you from being uniquely identifiable.

5. Your raw biometric data is never seen by the server (BP-005 — Homomorphic Processing)

Separation ratios — the mathematical comparisons that distinguish you from other players — are computed on **encrypted data**. The server performs calculations on ciphertext. It never possesses your raw biometric features. Even if the server were fully compromised by an attacker, they would only obtain encrypted data that cannot be decrypted without keys they do not have.

6. Your identity is split across 16 agents (BP-006 — Shamir Sharing)

Your biometric identity is not stored in one place. It is split into **16 shares** using Shamir's Secret Sharing scheme — you need **8 of 16 shares** to reconstruct your identity. A single compromised agent reveals nothing (one share is mathematically equivalent to random bytes). Up to **7 agents can be simultaneously compromised** and your identity remains completely safe. No single point of failure.

7. Raw data lives only in RAM and is securely destroyed (BP-007 — Ephemeral Sessions)

Your raw biometric data is **never written to disk**. It exists only in RAM during your play session, with a maximum window of 30 minutes. The RAM pages are locked using `mlock()` to prevent the operating system from swapping them to disk. When your session ends, the memory is securely overwritten using constant-time secure erase (`memset_secure()`). Only derived statistical thresholds and zero-knowledge proofs survive the session. The raw data is gone — completely, securely, irreversibly.

4.4 What Goes On-Chain (And What Doesn

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t)

VAPI uses the IoTeX blockchain for its public record-keeping. Here is the exact separation between public and private data:

ON-CHAIN (Public)	NEVER ON-CHAIN (Private)
Device eligibility status (true/false)	Your humanity_probability score
PoAC record hashes (NOT the records themselves)	Your raw biometric features
ZK proof verification results	Your personal biometric profile
Ruling history (BLOCK/CLEAR decisions)	Your calibration data
Credential status (active/expired/suspended)	Your individual L4 Mahalanobis centroid

The on-chain data proves you're eligible. The private data stays private. **This separation is an immutable protocol guarantee** — it cannot be changed by any software update, patch, or governance decision. The architecture physically prevents private biometric data from reaching the blockchain.

5. Enrollment

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What It Means and What It Costs You

5.1 The Enrollment Process

Enrollment is the process by which VAPI builds your personal biometric profile and issues your credential. It is straightforward:

1. **Play 10 normal sessions** with the VAPI bridge running and your controller connected via USB
2. During these sessions, VAPI records your controller physics and builds your biometric profile — your unique tremor frequency, trigger behavior, grip pattern, timing characteristics
3. Each session must be classified as **NOMINAL** — meaning VAPI confirmed you were playing normally as a human, with no cheat codes or major anomalies
4. You do not need to do anything special. No calibration exercises, no tests, no specific games. Just play with your controller connected to the VAPI bridge.
5. After 10 clean NOMINAL sessions, your **PHG (Player Human-Genome) credential** is issued as a soulbound token on the IoTeX blockchain

5.2 The Soulbound Credential (VHP)

Your **Verified Human Proof (VHP)** credential is a soulbound ERC-4671 token with specific properties designed to kill identity fraud at the cryptographic level:

- It **cannot be transferred, sold, lent, or traded**. It is permanently bound to the wallet associated with your biometric profile.
- It is **cryptographically bound to YOUR biometric profile and YOUR device**. Another person using your account will fail L4 biometric verification.

- It has an **expiration date** linked to the 90-day biometric decay cycle. You need to stay active to keep it valid.
- If your credential expires, you re-enroll by playing additional sessions. If your biometric profile drifts significantly (persona break detection), you may need re-enrollment with attestation.

What this kills: Nobody can buy your credential. Nobody can borrow it. Nobody can use your account and have YOUR biometric proof attached. Account boosting becomes cryptographically impossible — a booster playing on your account will produce a different biometric signature, and the L4 layer will detect the mismatch.

5.3 What Enrollment Costs

Cost Category	Details
Monetary cost	Currently: **none**. The VAPI bridge software is free to run on your PC.
Hardware requirement	A compatible controller. The reference device is the **Sony DualShock Edge (CFI-ZCP1)** for full Attested tier (L0–L6). Other controllers may work at Standard tier (L0–L5) as they get certified.
Time investment	10 gameplay sessions. Each session is a normal play session — no special duration requirements, no tests.
Privacy trade-off	You are sharing controller physics data with the VAPI bridge running locally on your PC. Raw data never leaves your machine (BP-007). Only derived statistical thresholds are used for your biometric profile.

6. Device Tiers

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What Your Controller Gets You

Not all controllers are equal in the VAPI ecosystem. The detection capabilities available to you depend on your controller's hardware and how it connects to your PC.

6.1 Attested Tier (Highest Trust)

- **All 9 detection layers active** (L0 through L6)
- USB 1000 Hz transport required
- Controller must be **PHCI-certified** (Physical Hardware Certification Index) — verified by the protocol to have the sensors and secure element necessary for full biometric trust
- Full biometric stack including adaptive trigger resistance measurement via L6 haptic challenge (when enabled)
- Currently: **Sony DualShock Edge (CFI-ZCP1)** is the reference Attested device
- **What this means for you:** Maximum protection, full tournament eligibility, strongest possible proof of human play

6.2 Standard Tier

- **L0 through L5 minimum** (L4 may be partial — gyro only, no touchpad features; L6 optional)
- BLE 250 Hz transport acceptable (when enabled — see Section 7.4)

- Controller does not need PHCI certification (self-registered)
- **What this means for you:** Strong protection and tournament eligibility, but some detection layers may operate with reduced precision compared to Attested tier. Hardware injection detection (L2) and behavioral cheat detection (L3) are fully functional.

6.3 Fallback / Unverified

- **L0 through L3 only** (structural integrity + hard cheat detection)
- No biometric verification — no L4, no L5, no L6
- **Never eligible for tournament anti-cheat**
- Useful only for casual analytics and basic cheat screening
- **What this means for you:** Basic cheat detection but no identity verification. You cannot compete in VAPI-gated tournaments at this tier.

Feature	Attested	Standard	Fallback
Hardware injection detection (L2)	✓	✓	✓
Behavioral cheat detection (L3)	✓	✓	✓
Biometric fingerprint (L4)	✓ Full	Partial	✗
Temporal rhythm (L5)	✓	✓	✗
Haptic challenge (L6)	✓ (when enabled)	Optional	✗
Tournament eligibility	**Full**	**Yes**	**No**
VHP credential	✓	✓	✗

7. What VAPI Cannot Do Yet

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Honest Limitations

⚠ **TRANSPARENCY NOTICE**

This section exists because you deserve to know what is not ready. Every limitation listed here is tracked publicly in the VAPI invariants. None of this is hidden. If you are evaluating VAPI for your tournament, your team, or your community, this section is required reading.

7.1 Separation Ratio

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Still in Development

The **separation ratio** measures how well VAPI can distinguish between different players biometrically — it is the core metric for identity verification (L4).

Metric	Status
Current separation ratio	**0.728** (touchpad_corners probe, N=35 sessions, 3 players)
Target for tournament deployment	**>1.0**
Path forward	More calibration sessions, more enrolled players, resting tremor probe technique expected to push ratio above 1.0

What this means in practice: VAPI can detect hardware cheats (L2) and behavioral cheats (L3, L5) **right now**. The biometric identity verification — proving it is specifically *you* and not someone else — is promising but needs more calibration data. At 0.728, the system can often distinguish between players, but not reliably enough for tournament-grade identity assertions. The protocol is transparent about this. The separation ratio is tracked publicly in the VAPI invariants, not hidden behind marketing language.

7.2 L6 Active Challenge

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Disabled

L6 (haptic challenge-response) is currently disabled. It requires N≥50 calibration sessions to activate safely — the system needs enough data about how different humans respond to sub-perceptual vibration before it can reliably distinguish human reflexes from non-response.

When active, L6 would measure your body's involuntary reflex response to haptic stimuli you cannot consciously perceive. Currently, L6 contributes **nothing** to your humanity_probability score. The formula runs without it.

7.3 GSR (Galvanic Skin Response)

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Hardware Not Yet Available

GSR measures emotional arousal through changes in skin conductance — your palms sweat slightly more under competitive stress, and this response correlates with in-game events for human players (Pearson r > 0.3 vs game events) while bots show no correlation (r ≈ 0.0).

This feature requires an **ESP32-S3 BLE grip accessory** that is not yet manufactured. It would add a future L7 layer with physiological biometric data. **Timeline:** Future phase, pending hardware development. No date has been set because the hardware does not exist yet.

7.4 Bluetooth Transport

BLE 250 Hz is defined in the architecture as a fallback transport, but it is **currently disabled** (`bt_transport_enabled=false`). USB 1000 Hz is the only active transport. BLE thresholds require separate calibration with N≥50 sessions, independent of USB calibration, and this has not been completed.

7.5 Dry Run Mode

> ⚠ **IMPORTANT**
>
> The system currently operates in `dry_run=true` mode — enforcement simulation. This means BLOCK rulings are **recorded but not actually enforced**. No real penalties are applied to any player.

VAPI will exit dry run mode only after **N≥100 live adjudications with zero false positives**. This is a deliberate safety measure. The team will not apply real penalties to real players until the system has proven it does not produce false positives under live conditions. This protects you from being unfairly punished during the calibration phase.

7.6 Limited Controller Support

Currently, only the **Sony DualShock Edge (CFI-ZCP1)** is fully supported as the reference Attested device. Other controllers — Xbox Elite V2, Xbox Standard, Switch Pro Controller — are defined in the architecture and have preliminary specifications, but each requires per-controller calibration of biometric thresholds because different controllers have different physical characteristics (weight, button travel, stick resistance, IMU sensitivity).

Multi-controller support is disabled by default (`multi_controller_enabled=false`). As more controllers are individually calibrated and certified, the ecosystem will expand.

7.7 Contract Status

All **43 smart contracts** are live on **IoTeX testnet** (chain ID 4690), **not mainnet**. This means the entire on-chain infrastructure is functional but running on test infrastructure.

Mainnet migration requires four prerequisites to be met:

6. Security audit of all 43 contracts by an independent auditor
7. Separation ratio confirmed >1.0 with sufficient sample size
8. 100 live validated adjudications with zero false positives
9. VAPIToken TGE demonstrated end-to-end on testnet

Until all four are complete, VAPI operates on test infrastructure. This is a deliberate sequencing decision — no deployment to production until the technology meets its own stated requirements.

8. The VAPI Token

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What It Means For Gamers

8.1 Earning Potential

VAPI tokens are earned through **participation, not purchase**. Active play with PoAC submissions generates a base earning rate, which can be increased through multipliers:

Activity / Status	Multiplier
Base rate (active play with PoAC)	1.0×
Tournament Passport holder	**1.5×**
Enrolled (biometric credential active)	**2.0×**
CERTIFY streak (5+ consecutive clean sessions)	**2.5×**
MPC ceremony contributor	**1.25×**
Gate attestation anchor	**3.0×**

These multipliers reward consistent, verified human play. They incentivize exactly the behavior that benefits the ecosystem — playing honestly, building a track record, and contributing to the protocol's calibration data.

8.2 Data Sovereignty

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You Own Your Data

The **DataSovereigntyRegistry** smart contract gives you explicit control over what happens with your anonymized biometric data. You can pledge data sovereignty at different tiers — or not at all:

Tier	What It Allows
GAMER	Your anonymized data can be used for competitive analytics only (matchmaking improvement, anti-cheat calibration)
DEVELOPER	Your anonymized data can be licensed for game AI training (improving NPC behavior, bot detection models)
MANUFACTURER	Your anonymized data can be used for hardware improvement (controller ergonomics, sensor calibration)
None	Your data is used only for your own anti-cheat verification. Nothing else.

If you opt in, your anonymized biometric data can be licensed through the **VAPIDataMarketplace** contract. Revenue split: **70%** goes to your device pool / **30%** to protocol treasury. All data flows are consent-gated via ZK-attested consent (BP-002) — nothing happens without your cryptographic consent, and you can revoke at any time.

8.3 Token Launch — Not Until It's Ready

The token launch is sequenced behind **measurable milestones**, not arbitrary dates or market timing:

10. Inter-person separation ratio **>1.0** confirmed empirically with sufficient sample size
11. **N≥100** live non-dry-run adjudications with **zero false positives**

12. VHP end-to-end flow demonstrated on IoTeX testnet
13. All 43 contracts independently audited

Only after all four milestones are met will TGE (Token Generation Event) be considered.

ⓘ **HONEST NOTE**

VAPI will not launch a token until the technology actually works at the level it promises. This is deliberately anti-hype. There is no “coming soon” date, no countdown timer, no pre-sale. The milestones are public, measurable, and non-negotiable.

9. How VAPI Changes Competitive Gaming

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The Real Impact

9.1 The Level Playing Field

When VAPI is fully integrated into a competitive environment, the fundamental dynamics of online competition change:

- **Every participant has cryptographic proof of human presence** — not a software scan, not a trust assumption, but a physics-verified chain of evidence
- **Hardware injection devices are detectable for the first time in gaming history** — the Cronus Zen problem that has plagued competitive gaming for years has a solution

- **Account boosting is cryptographically impossible** — soulbound credentials bound to individual biometric profiles mean a different player produces different physics
- **Rankings reflect actual player skill** because identity fraud is verifiable, not just suspected
- **Prize pools are protected by on-chain audit trails** — every eligibility decision has a cryptographic record

9.2 The Player Dashboard

VAPI provides full transparency through the player dashboard — the **MY CONTROLLER** page:

- A live 3D digital twin of your controller showing real-time biometric heartbeat visualization
- Real-time display of your biometric status, humanity score, and PoAC chain health
- You can see exactly what VAPI is detecting at every moment — **full transparency, not a black box**
- Your credential status, device tier, enrollment progress, and complete session history are visible
- If there is an anomaly or flag, you can see it and understand why it was generated

This is a deliberate design decision. VAPI believes that a system you cannot inspect is a system you should not trust. The dashboard gives you complete visibility into what the system sees and what conclusions it draws.

9.3 Tournament Integration

Tournaments using VAPI run a single eligibility check:

```
isFullyEligible(deviceId)

→

true / false
```

This one call verifies all of the following simultaneously:

- Device registration status
- ZK session proof validity
- Biometric credential (VHP) validity and expiration
- Ruling history is clean (no BLOCK rulings)
- On-chain anchoring is present and intact

No new software installations required on tournament day. No invasive system scans. No kernel drivers to install. No "please restart your PC" delays. Just: "Are you verified?" → yes or no, with cryptographic proof backing the answer.

9.4 What It Means for Ranked Play

VAPI-gated ranked playlists create an environment where:

- Every opponent in the lobby has **proven they are a real human** playing with their own controller
- Smurfs cannot hide — their biometric profile differs from the account owner's profile, and L4 detects the discrepancy
- Hardware cheat devices are detected in **real-time**, not weeks later after manual review

- Your rank means what it is supposed to mean — a reflection of your skill against other verified humans

This does not eliminate every form of cheating or every competitive integrity issue. What it does is eliminate the categories of cheating that current systems are structurally incapable of addressing — hardware injection, identity fraud, and macro automation through hardware relay.

10. Frequently Asked Questions

Q: Can VAPI identify me personally from my controller data?

A: No. VAPI’s biometric features are statistical patterns derived from controller physics — tremor frequency, grip pressure distribution, timing jitter. These features distinguish you from other players for identity verification purposes within the system, but they cannot identify you as a real-world individual. Your data is protected by seven privacy primitives: automatic 90-day decay (BP-001), K-anonymity cohorts of 5+ players (BP-004), ephemeral RAM-only storage with secure deletion (BP-007), Shamir sharing across 16 agents (BP-006), homomorphic processing on encrypted data (BP-005), differential privacy with $\varepsilon \leq 1.0$ (BP-003), and ZK-attested consent (BP-002).

Q: What if I’m falsely flagged?

A: Hard cheat codes (`0x28`, `0x29`, `0x2A`) only fire when the physics are definitive. For example, L2 hardware injection detection requires the IMU data to physically contradict the input data — a button press with no corresponding controller movement. This is not a probability judgment; it is a physics measurement. Advisory codes contribute to humanity_probability but a single advisory flag never triggers a block. Full adjudication requires consensus from 36 independent agents AND a decentralized network with a 0.65 epistemic threshold designed so no single agent or pair of agents can reach it alone. Additionally, the system is in `dry_run` mode — no actual penalties are applied while calibration continues.

Q: Does VAPI slow down my game?

A: No. The VAPI bridge runs as a separate process on your PC. It reads controller data via USB HID at 1,000 Hz — the same data your game already reads. VAPI does not inject anything into your game process, does not hook your graphics pipeline, does not modify any game memory. Your game runs exactly as before. The bridge's processing is independent and does not compete with your game for GPU or critical CPU resources.

Q: What if I break my controller and get a new one?

A: Your biometric profile is tied to your identity, not to a specific physical controller — but your device registration will need to be updated. If you switch to the same model (DualShock Edge to DualShock Edge), re-enrollment is straightforward because the physical characteristics are similar. If you switch to a different controller model, you will need new calibration sessions because different controllers have different weight, button travel, stick resistance, and IMU sensitivity — all of which affect the biometric readings.

Q: Can someone clone my PoAC chain to impersonate me?

A: No. Every PoAC record is signed with ECDSA-P256 using the private key stored in your controller's secure element. Without physical possession of your specific controller (and its hardware secure element), the cryptographic signature cannot be generated. Additionally, the chain is linked via SHA-256 hashes — inserting a fake record breaks the chain integrity, which L1 immediately detects.

Q: What happens if VAPI detects someone using a Cronus Zen in my lobby?

A: If L2 fires (inference code `0x28`), the player's session is flagged as `DRIVER_INJECT`. In a VAPI-gated tournament, this blocks eligibility unconditionally. In the current `dry_run` mode, the detection is logged but no live penalty is applied. When enforcement goes live (after 100 adjudications with zero false positives), hardware injection detection will result in immediate session blocking and credential suspension.

Q: Is VAPI open source?

A: VAPI publishes its invariants, circuit specifications, and corpus methodology publicly. The protocol is designed for transparency — every eligibility decision has an on-chain audit trail. Specific implementation details of the bridge software and agent fleet are proprietary, but the cryptographic guarantees are publicly verifiable. You can independently verify that a PoAC chain is intact, that a credential is valid, and that a ruling was correctly computed.

Q: What if I play on multiple platforms or systems?

A: Your VHP credential is tied to your device identity, not a specific gaming platform or console. As long as your VAPI bridge is running on your PC and your controller is connected via USB, the credential works across any game or platform that integrates VAPI. The verification is at the controller and biometric level, not the application level.

11. Getting Started

—

Your First Steps

If you have read this guide and want to proceed, here is exactly what to do:

Step	Action	Details
1	Get a compatible controller	**Sony DualShock Edge (CFI-ZCP1)** for full Attested tier with all detection layers. Other controllers may work at Standard tier as they are certified.
2	Connect via USB	Direct USB connection. 1,000 Hz transport. No adapters, no hubs, no wireless dongles. Direct

Step	Action	Details
		cable from controller to PC.
3	Install the VAPI bridge	Lightweight application that runs in user space on your PC. No kernel drivers. No admin privileges required for operation.
4	Play 10 sessions	Normal gameplay. Nothing special required. The bridge builds your biometric profile automatically in the background. Each session must be classified as NOMINAL.
5	Receive your credential	After 10 NOMINAL sessions, your PHG credential is issued as a soulbound ERC-4671 token on the IoTeX blockchain.
6	Check your dashboard	The MY CONTROLLER page shows your live biometric status, PoAC chain health, credential status, and session history.
7	Compete	Join VAPI-gated tournaments and ranked playlists with cryptographic proof of your humanity backing every session.

12. The Bottom Line

Here is what VAPI is, stripped of any embellishment:

VAPI is the first system that can detect hardware injection devices like Cronus Zen — the single biggest unsolved cheating problem in competitive gaming. No other anti-cheat system can do this because they all operate at the software layer, and hardware injection operates below it. VAPI reads controller physics, and physics does not lie.

It proves you are human, proves you are YOU, and creates an unbreakable cryptographic chain of evidence for every session you play. Every millisecond of gameplay is signed, hashed, chained, and verifiable.

Your biometric data is protected by seven novel privacy primitives that exceed any current regulatory standard — GDPR, CCPA, BIPA, EU AI Act. No raw biometrics ever leave your device. Your data expires automatically on a 90-day half-life. Your identity is split across 16 independent agents using Shamir’s Secret Sharing. Your raw data lives only in locked RAM and is securely destroyed when your session ends.

⚠ WHAT’S NOT READY YET

The separation ratio is at **0.728** — it needs to reach **>1.0** before biometric identity verification is tournament-grade. L6 haptic challenge is disabled. The system is in `dry_run` mode — no real penalties are applied. Only one controller model is fully supported. All contracts are on testnet, not mainnet. The token has not launched and will not launch until measurable milestones are met. This is honest engineering, not premature deployment.

When fully deployed, VAPI changes the fundamental equation of competitive gaming: every match, every tournament, every ranked session has cryptographic proof that real humans are playing with their own hands on their own controllers. Hardware cheats are detectable. Identity fraud is verifiable. Rankings mean what they are supposed to mean.

You own your data. You control what happens with it — full consent, full revocation, full transparency. And for the first time in gaming history, your skill has a mathematically verifiable proof.

That is what VAPI does. No more, no less. The technology is real, the limitations are disclosed, and the path forward is publicly tracked. Now you have the complete picture to decide for yourself.

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